

BT101

Introductory Biology

(Lecture 8)

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Introduction



Evolution of life:

- ❑ Origin of Life
- ❑ Darwin's concepts of evolution
- ❑ Biodiversity

Cell, the structural and functional unit of life:

- ❑ Three domains of life
- ❑ Cell types, cell organelles and structure
- ❑ Basic biomolecules of cell

Introduction



- ❑ **Evolution** is the change in the inherited characteristics of biological populations over successive generations.

OR Simply

- ❑ **Evolution** is the change in the genetic composition of a population from generation to generation.

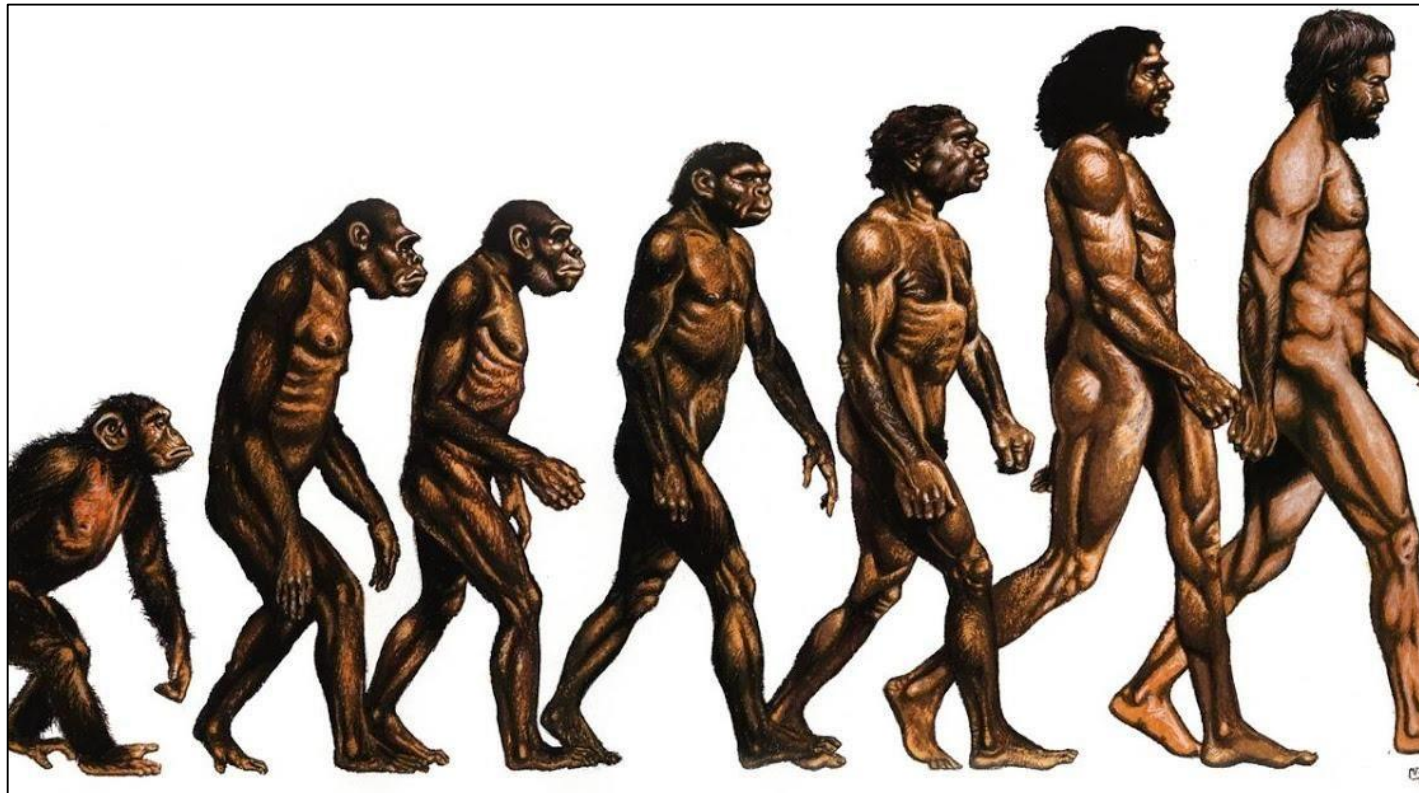
OR Simply

- ❑ **Evolution** is a process of gradual change that takes place over many generations, during which species of animals, plants, or insects slowly change some of their physical characteristics.

- ❑ **Evolutionary process** gives rise to diversity at every level of biological organization, including species, individual organism and molecules such as DNA and proteins.

Darwinism

- ❑ Darwinism is a theory of biological evolution.
- ❑ Darwinism is a theory of biological evolution by natural selection.
- ❑ Darwin defined evolution as "descent with modification," the idea that species change over time, give rise to new species, and share a common ancestor.



Darwinism



- ❑ It states that all species of organisms arise and develop through the process of natural selection of small, inherited variations that increase the individual's ability to compete, survive and reproduce. Also called Darwinian theory.
- ❑ The mechanism that Darwin proposed for evolution is natural selection. Because resources are limited in nature, organisms with heritable traits that favor survival and reproduction will tend to leave more offspring than their peers, causing the traits to increase in frequency over generations.
- ❑ Natural selection causes populations to become adapted, or increasingly well-suited, to their environments over time. Natural selection depends on the environment and requires existing heritable variation in a group.

Introduction

▼ **Figure 22.5 The voyage of HMS *Beagle* (December 1831–October 1836).**

Darwin in 1840,
after his return
from the
voyage



HMS *Beagle* in port

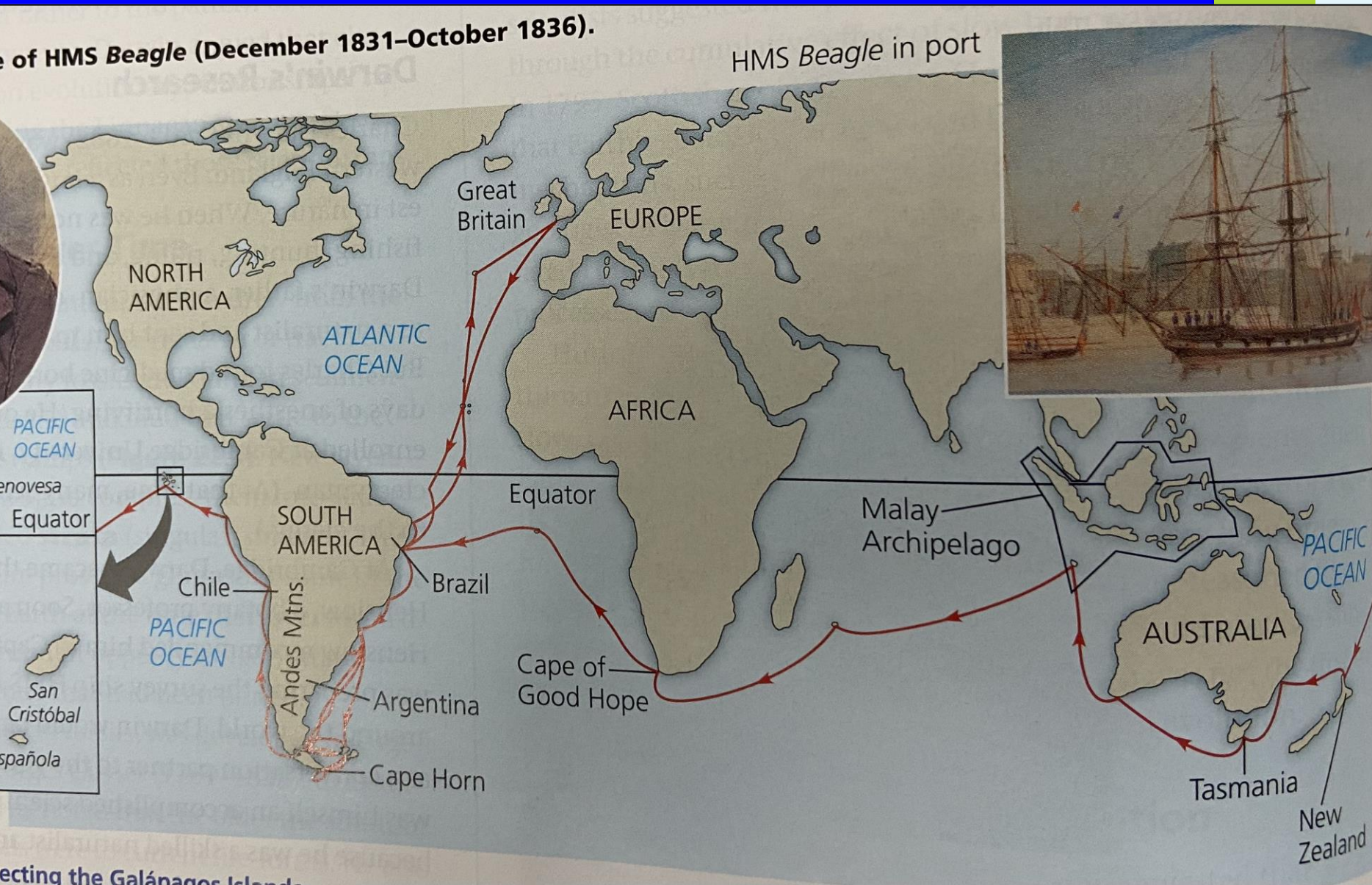
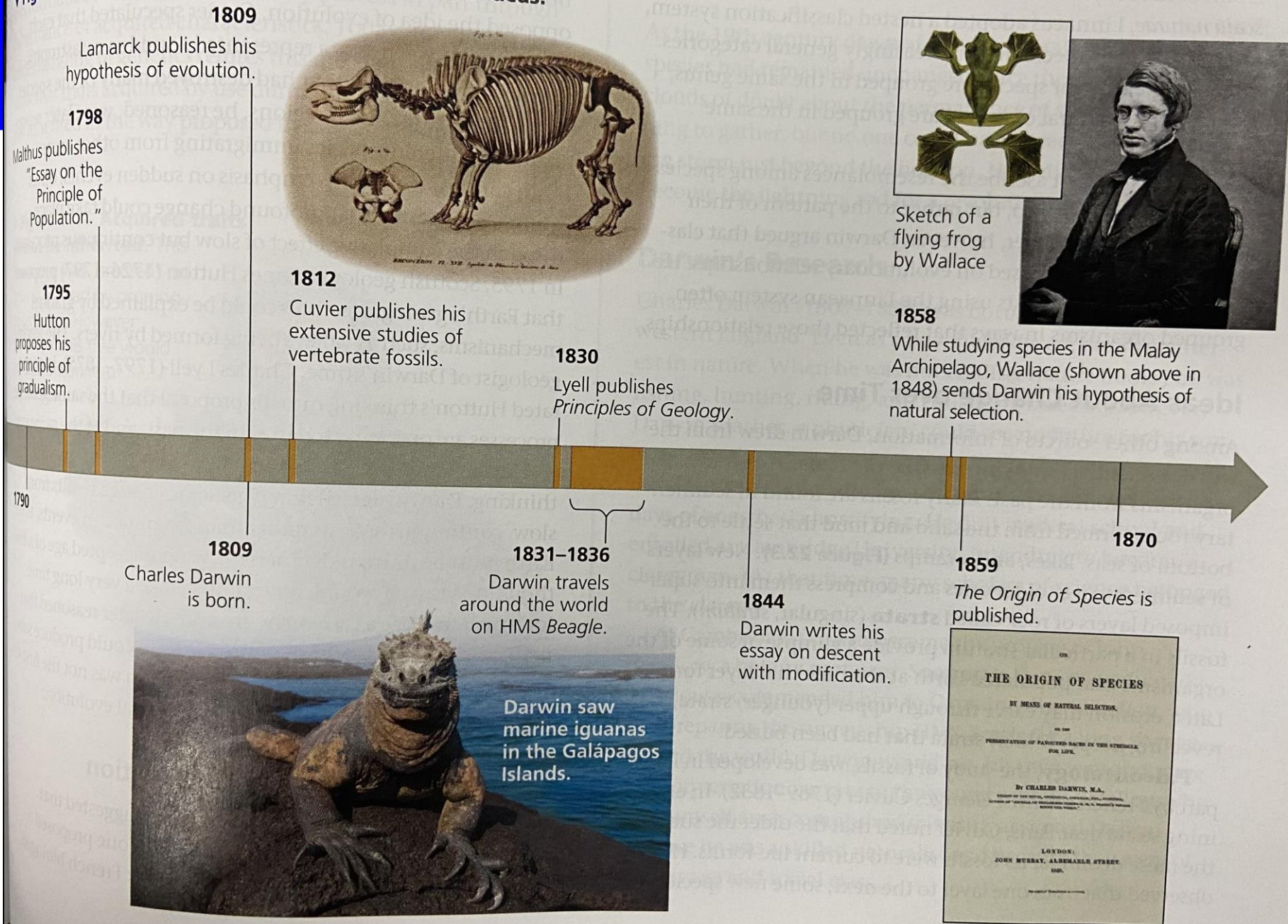


Figure 22.2 The intellectual context of Darwin's ideas.



Lamarckism



- ❑ In the early nineteenth century, the French naturalist **Jean-Baptiste Lamarck** published a book proposing that evolution took place and suggesting a mechanism by which it might occur.
- ❑ **Lamarck proposed that** modifications in an individual caused by its environment, or the use or disuse of a structure during its lifetime, could be inherited by its offspring and lead to a change in a species.
- ❑ **For instance, Lamarck suggested that giraffes have long necks because some of them stretched their necks during their lifetimes, then passed the stretched neck on to their children.**
- ❑ This mechanism for evolutionary change did not turn out to be correct, and today, we often remember Lamarck as the one who "got the wrong answer."
- ❑ However, before we judge, we should remember that Lamarck was a pioneer despite his incorrect idea.
- ❑ He was one of the very first to seriously engage with the idea of evolution and to actually suggest how it might take place.

Darwin's theory

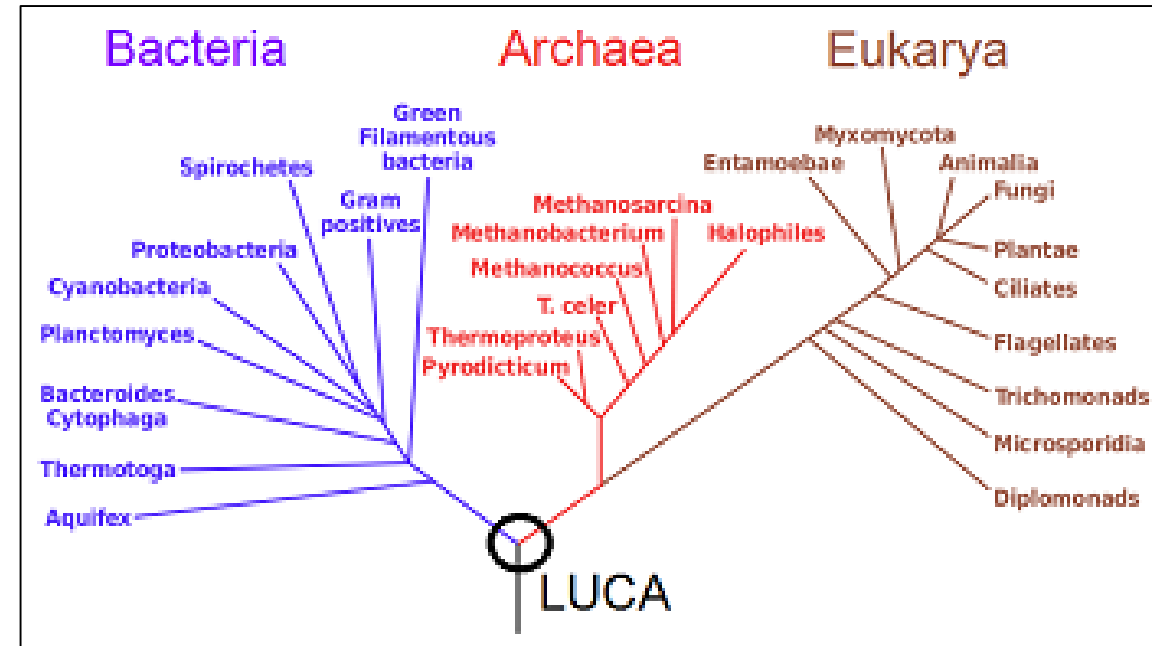


Four points of Theory

1. Common Descent
2. Gradualism
3. Multiplication
4. Natural Selection

1. Common Descent

- All organisms had **one common ancestor** in the past and with the change in climate and habitat, they **descended from that unknown ancestor**.
- Over a period of time, the changes within organisms got collected resulting in their survival for the changing environment. This is also termed as **adaptation**.

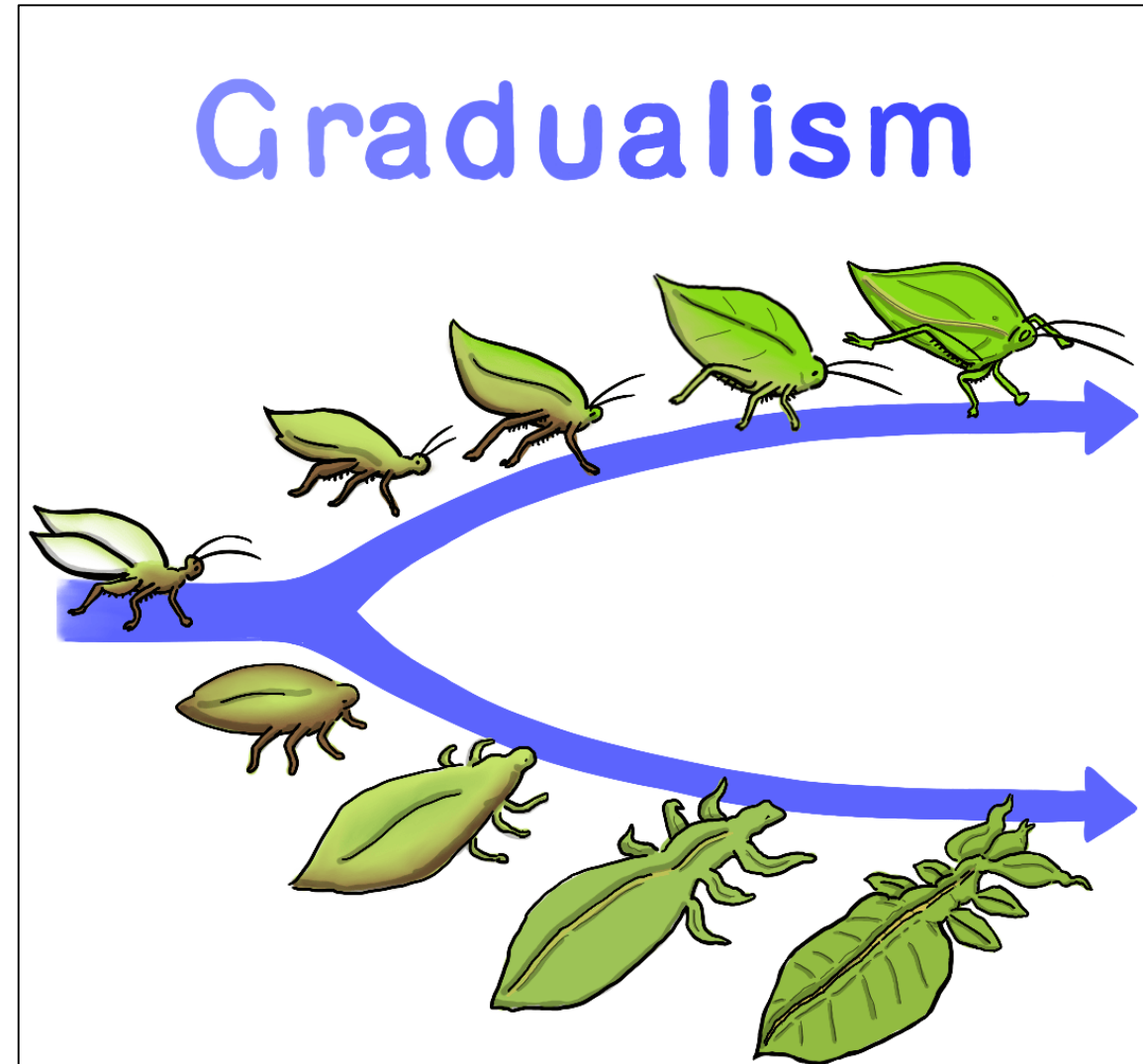




2. Gradualism

Origin and evolution of new species depend on the accumulation of changes. Over a period of time upon accumulating different changes, results in adaptation, and a new species is evolved. Hence, **evolution is a slow and gradual process.**

The different forms show the species slowly changing over many years.
***There are many generations between each form.**



3. Multiplication



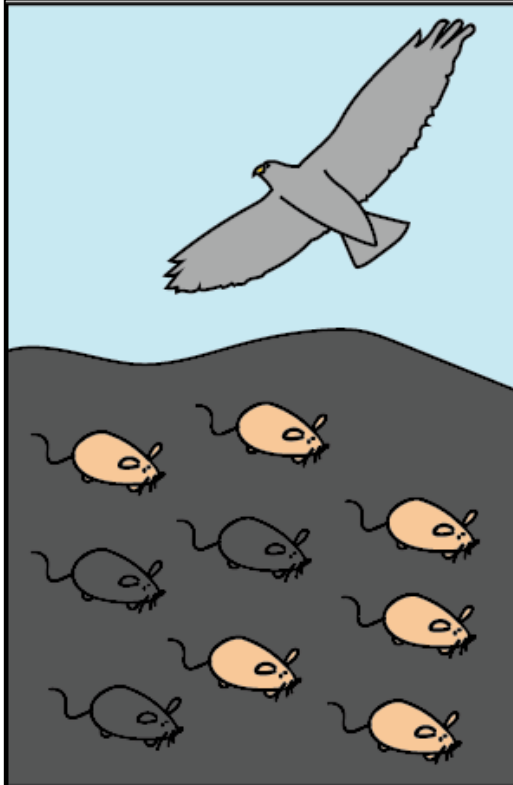
- Over a time period, changes with in species takes place.
- These changes can result in the species with similar in appearance or completely different in appearance and behavior.
- Hence, **species multiply** either having similar characteristics to their ancestor or different characteristics than them.

4. Natural Selection



- The evolutionary changes comes through the production of variation in each generation. The differential survival of individuals is based on different combinations of these variable characters.
- Individuals with characteristics which increase their probability of survival will have more opportunities to reproduce. **Their offsprings benefits from the inherited advantageous character which then spreads to their population.**

Natural Selection



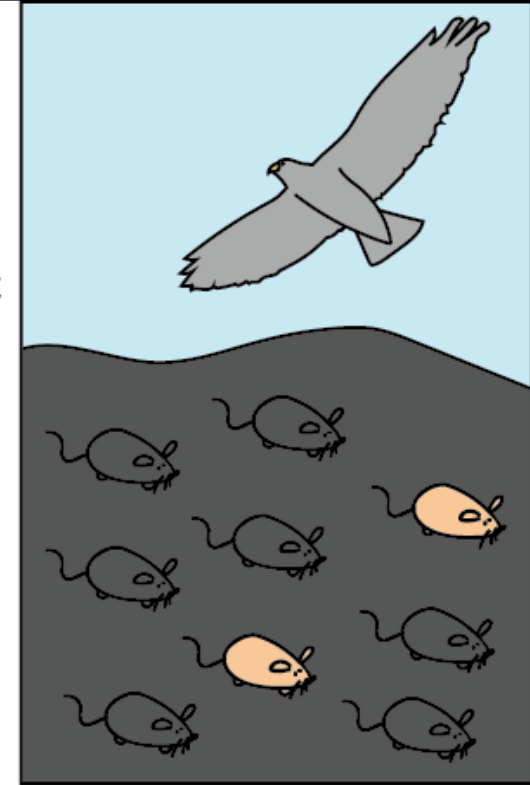
A population of mice has moved into a new area where the rocks are very dark. Due to natural genetic variation, some mice are black, while others are tan.

Some mice are eaten by birds



Tan mice are more visible to predatory birds than black mice. Thus, tan mice are eaten at higher frequency than black mice. Only the surviving mice reach reproductive age and leave offspring.

Mice reproduce, giving next generation



Because black mice had a higher chance of leaving offspring than tan mice, the next generation contains a higher fraction of black mice than the previous generation.

How Natural Selection occurs?

Natural Selection occurs in 5 steps



1. Overproduction

Every species on this planet has potential to give birth to young ones and increase its number. This is known as **Overproduction**. There is no limit for any species for the number of offsprings.



2. Competition

More the number of species, higher will be the competition for their survival. It is because, the environment has finite resources that can support the limited number of species.

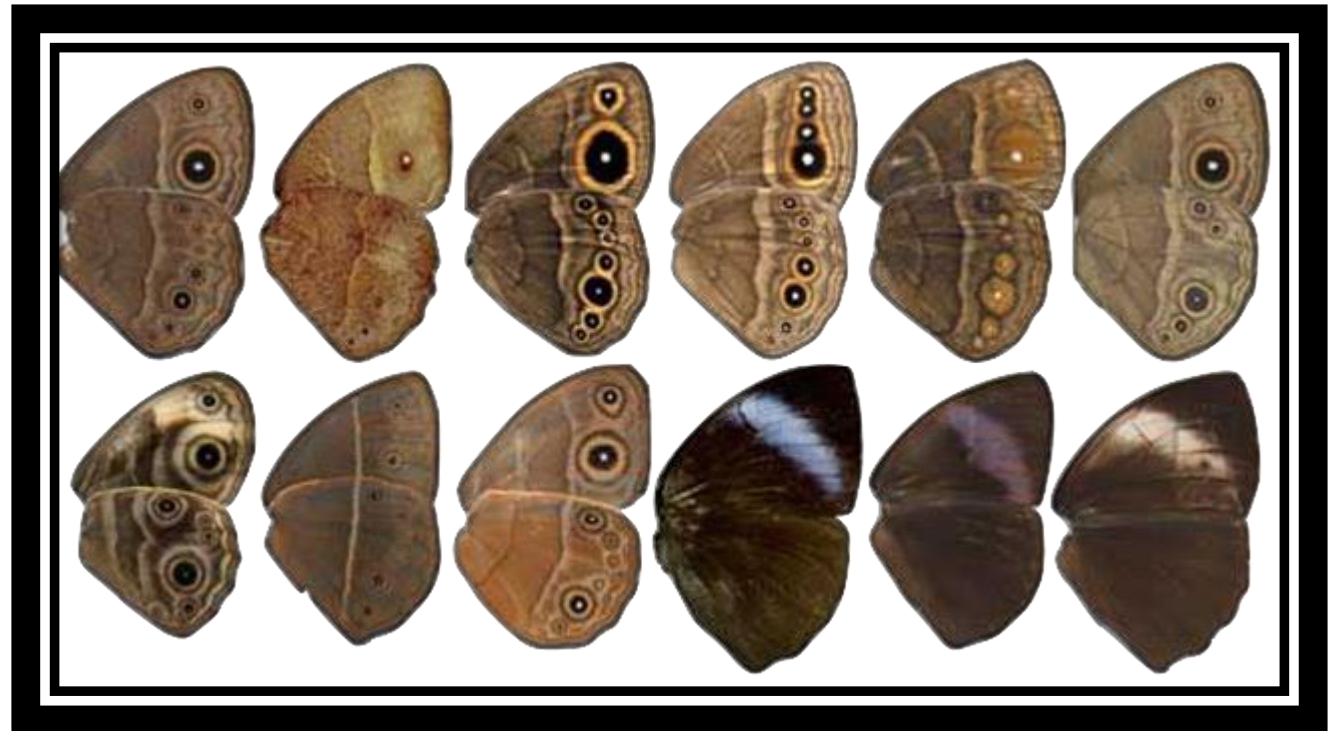
Hence, those offsprings who meet the environmental criteria can survive.



3. Variation

- Variation is the key ingredient for survival of the species.
- Those species which have the beneficial variations have higher chances of survival than those species without variation. Hence, as a result of variation, the species develop morphological and character differences.

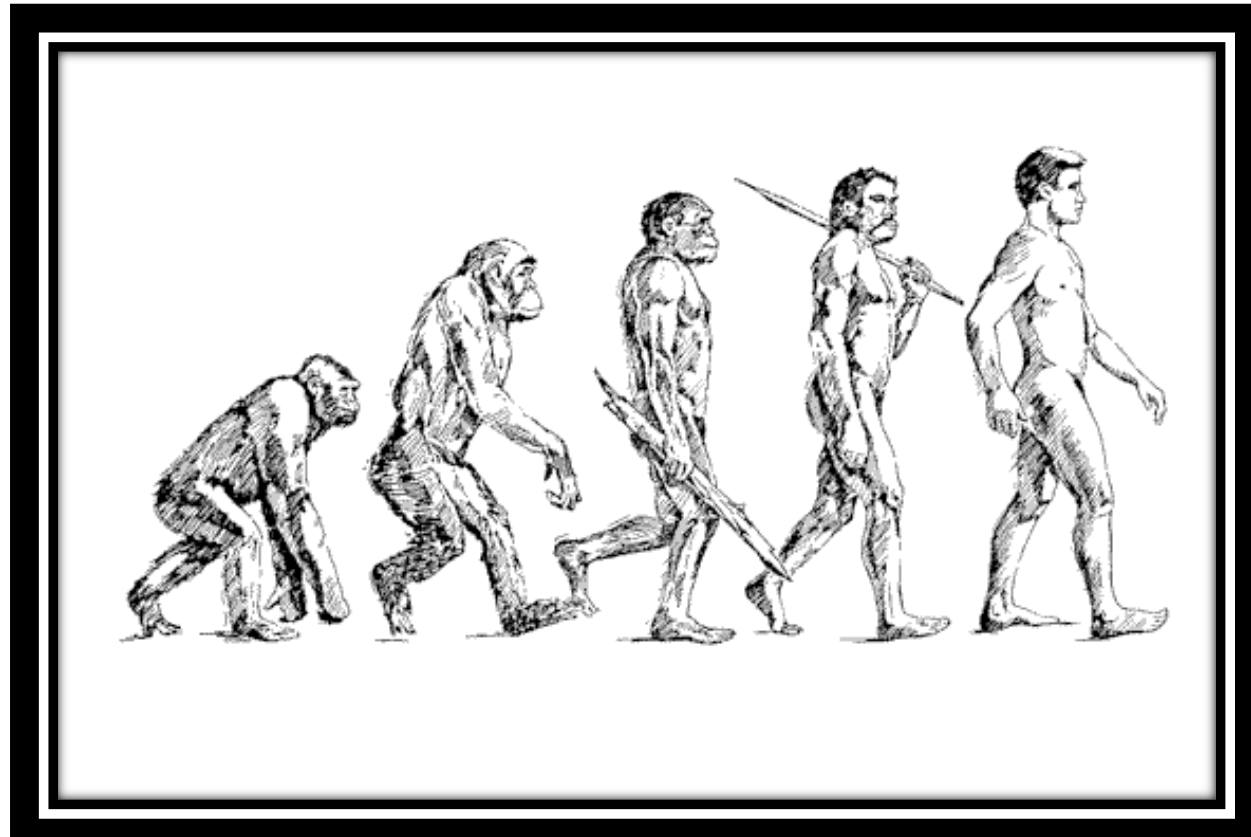
In predator-prey interaction, variation helps **butterflies having similar color wings as the dark background of tree bark** to hide from predators.



4. Survival of the fittest

Species which have developed the variation and can tackle the environmental conditions are fit for survival and reproduction. Hence, fitness depends upon the adaptation of species to a particular environment.

Therefore, any species with low fitness dies and that with higher fitness survives.

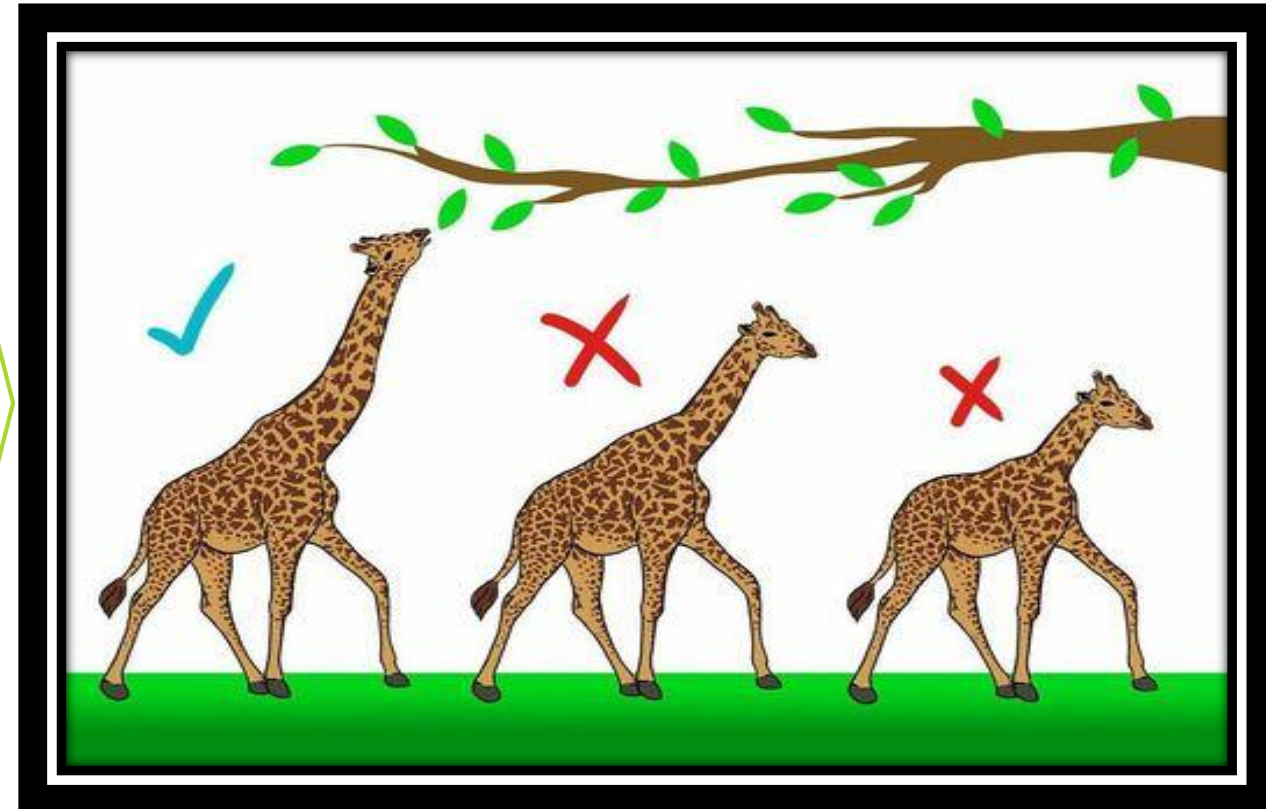


5. Natural selection

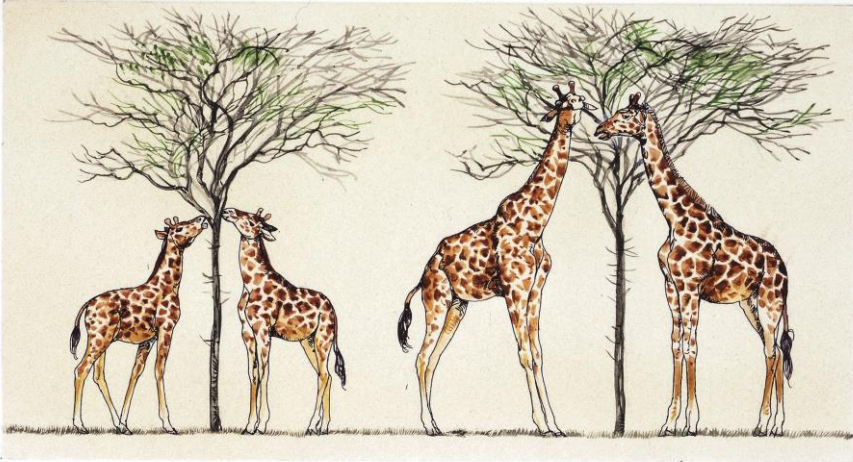
After a particular time period, the **changes or variation** acquired by the species results in the **Natural Selection**. This means that the nature selects the beneficial variation and increases the survival rate of the offspring.

According to **Darwin**, the long neck of Giraffe is the result of natural selection, acting through the animal's tree-feeding habit.

During dearth, the individuals who could reach even an inch or two above the others, survived.

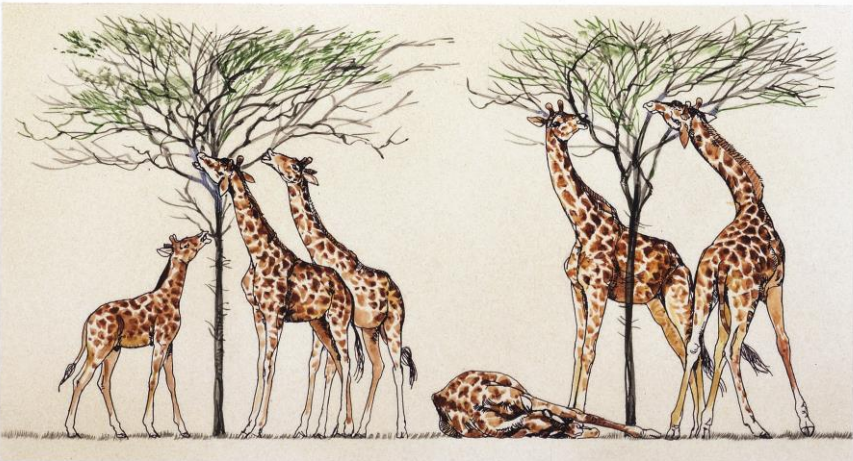


5. Natural selection



Theory of Jean-Baptiste Lamarck

and



Theory of Charles Darwin

Examples of Natural Selection

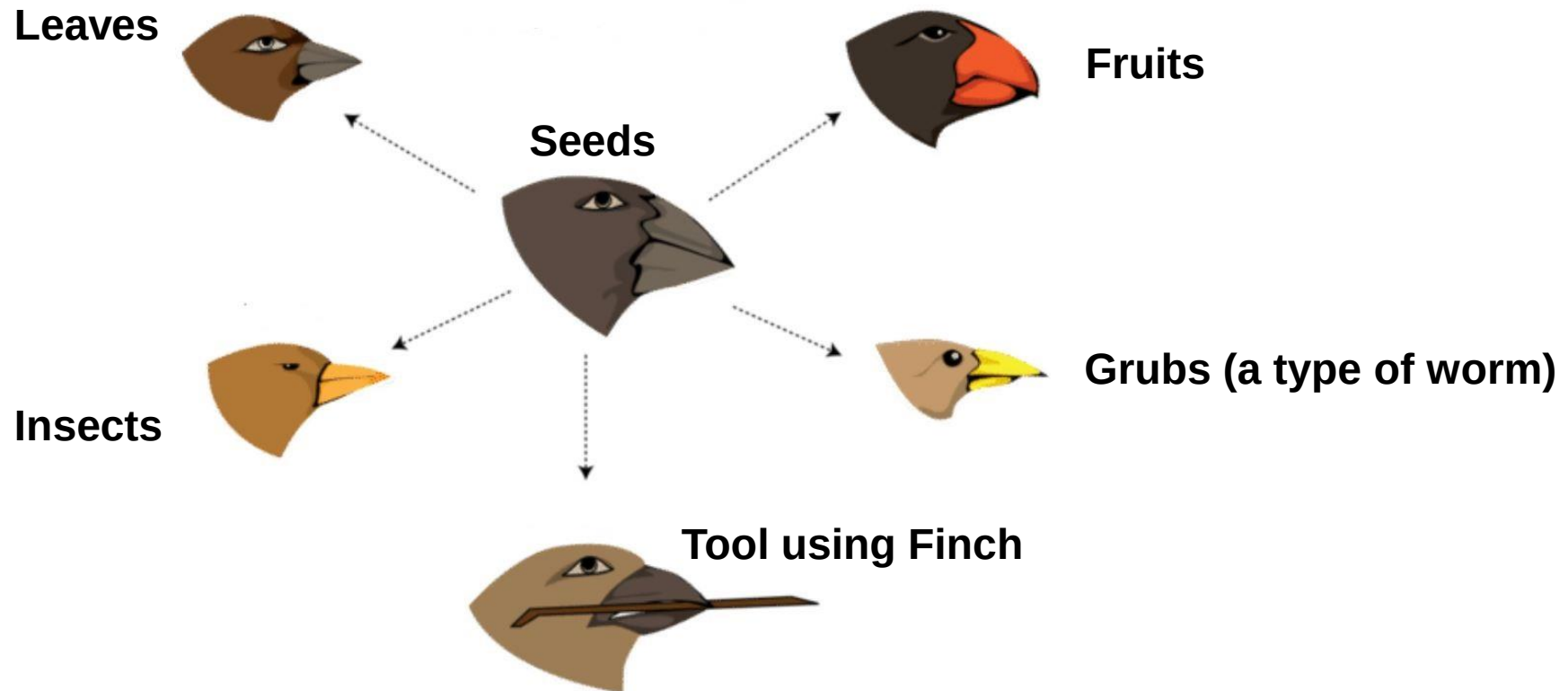


1. Drug Resistance in bacteria:

Antibiotic resistance is a consequence of **evolution** *via* natural selection. The **bacteria** which acquire mutation, allows them to survive and to reproduce. They pass this trait to their progeny, which will be a fully **drug resistant** generation.

Examples of Natural Selection

2. Shape of beak in finches: Darwin observed that the beak shape varies among Finch species. He postulated that the **beak** of an ancestral species had adapted over time to equip the **finches** for different food sources.



Example of Natural Selection

- 3. The Peppered Moth:** **Evolution** of **peppered moth** is an **evolutionary** instance of directional color change in their population as a consequence of Air Pollution during the Industrial Revolution. The number of dark-color **moths** increased at that time, which is an example of **industrial melanism**.

Light gray peppered moth (*Biston betularia*) and a dark pigmented variant rest near each other on the trunk of a soot-covered oak tree.

Against this background, the light gray moth is more easily noticed than the darker variant.



Video



Important Links to see Videos



- ❑ **Darwin's theory of evolution:**
- ❑ <https://www.youtube.com/watch?v=urSNtaQKOAk>
- ❑ <https://www.youtube.com/watch?v=0SCjhI86grU>